METRIC

MIL-L-21260D
AMENDMENT 2
11 October 1990
SUPERSEDING
AMENDMENT 1
23 May 1989

MILITARY SPECIFICATION

LUBRICATING OIL, INTERNAL COMBUSTION ENGINE, PRESERVATIVE AND BREAK-IN

This amendment forms a part of MIL-L-21260D, dated 29 April 1988, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 3

2.2, under AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM), D 2896, delete the word "Total" from the title and add:

"D 4624 - Measuring Apparent Viscosity by Capillary Viscometer at High Temperature and High-Shear Rates.
"D 4683 - Measuring Viscosity at High Temperature and High Shear Rate by Tapered Bearing Simulator.
"D 4741 - Measuring Viscosity at High Temperature and High Shear Rate by Tapered-Plug Viscometer.
"D 4927 - Elemental Analysis of Lubricant and Additive Components - Barium, Calcium, Phosphorus, Sulfur, and Zinc by Wavelength Spectroscopy."

Delete "DETROIT DIESEL ALLISON (DDA)" and substitute "ALLISON TRANSMISSION DIVISION (ATD)".41.5

AMSC N/A FSC 9150
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2.2, add the following at the end of the section:

"DEUTSCHES INSTITUT FUR NORMUNG (DIN)

DIN 51581 - Determination of evaporation loss of lubricating oils.

(Application for copies should be addressed to Deutsches Institut fur Normung e.V., Burggratenstr 4-10, D-1000 Berlin 30, Germany T 26011.)"

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Table II, before "Borderline pumping temperature, °C (max)" insert "High temperature/high shear Viscosity" and a "X" for all grades to indicate value shall be reported.

Under "Viscosity index, (min)", grade 40, delete the "X" and substitute "80".

After "Flash point, °C, min" insert "Evaporative loss" and a "X" in columns 10W and 15W/40 grades.

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3.4.5, after "Cam wear, μm, Average (max)" delete "196" and substitute "203" and after "Maximum" delete "441" and substitute "457".

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4.5.2, after line 16, add "High temperature/high shear" and "Evaporative loss".

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Table III, delete in its entirety and substitute the following:

TABLE III. Test methods.

Test	Test method			
	FED-STD-791	ASTM	SAE	
Viscosity, kinematic		D 445		
Viscosity, apparent 1/			J300	
High temperature/high shear		D 4683, D 4624, D 4741		
Viscosity index		D 2270		
Pour point		D 97		
Stable pour point	203			
Borderline pumping		D 4684		
Flash point		D 92		
Evaporative loss 2/		D 2887		51581
Gravity, API		D 287		
Carbon residue		D 524		
Color		D 1500		
Total acid number		D 664		
Base number		D 2896		
Phosphorus		D 1091, D 4047		
Chlorine		D 808 or D 1317 3/		
Sulfur 4/		D 129, D 1552, D 2622, D 4294		
Nitrogen		D 3228		
Saponification number		D 94		
Sulfated residue		D 874		
Boiling range distribution		D 2887		
Metallic components	5601	D 4628 5/, D 4927		
Foaming		D 892		
Stability & compatibility	3470 6/			
Moisture-corrosion characteristics		Sequence IID 7/		
Oxidation & wear characteristics		Sequence IIIE 7/		
Low temperature deposits & wear		Sequence VE 7/		
Bearing corrosion & shear stability		Labeco L-38 8/		
Ring-sticking, wear, & accumulations of deposits:				
Four-stroke cycle diesel engine 8/		Caterpillar 1G2		
Two-stroke cycle diesel engine	355			
Friction retention characteristic & wear:				
Slip time & wear		Allison C-3 9/		
Stopping time & wear		D 4736		
Seal compatibility		Allison C-3 10/		

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1/ Obtain the apparent viscosity using the method of test set forth by appendix A of SAE J300.
2/ The DIN 51581 is the preferred method.
3/ ASTM D 808 is the preferred method.
4/ ASTM D 1552 is the preferred method. ASTM D 4294 is only for use with base stocks.
5/ Other spectrochemical analysis methods as approved by the qualifying activity (see 6.4) may be used as alternates.
6/ See 4.6.1 for clarifying instructions.
7/ In accordance with ASTM STP 315H.
8/ In accordance with ASTM STP 509A.
9/ Use procedure described in item 9, Allison Transmission Division (ATD) C-3 specification.
10/ Use procedure described in item 6, Allison Transmission Division (ATD) C-3 specification.

Custodians:
Army - ME
Navy - Sh
Air Force - 68

Preparing activity:
Army - ME

Project 9150-1072

Review activities:
Army - AR, SM
Navy - AS, MC, SA, YD
Air Force - 11
DLA - GS

User activities:
Army - AT, MI
Navy - OS

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